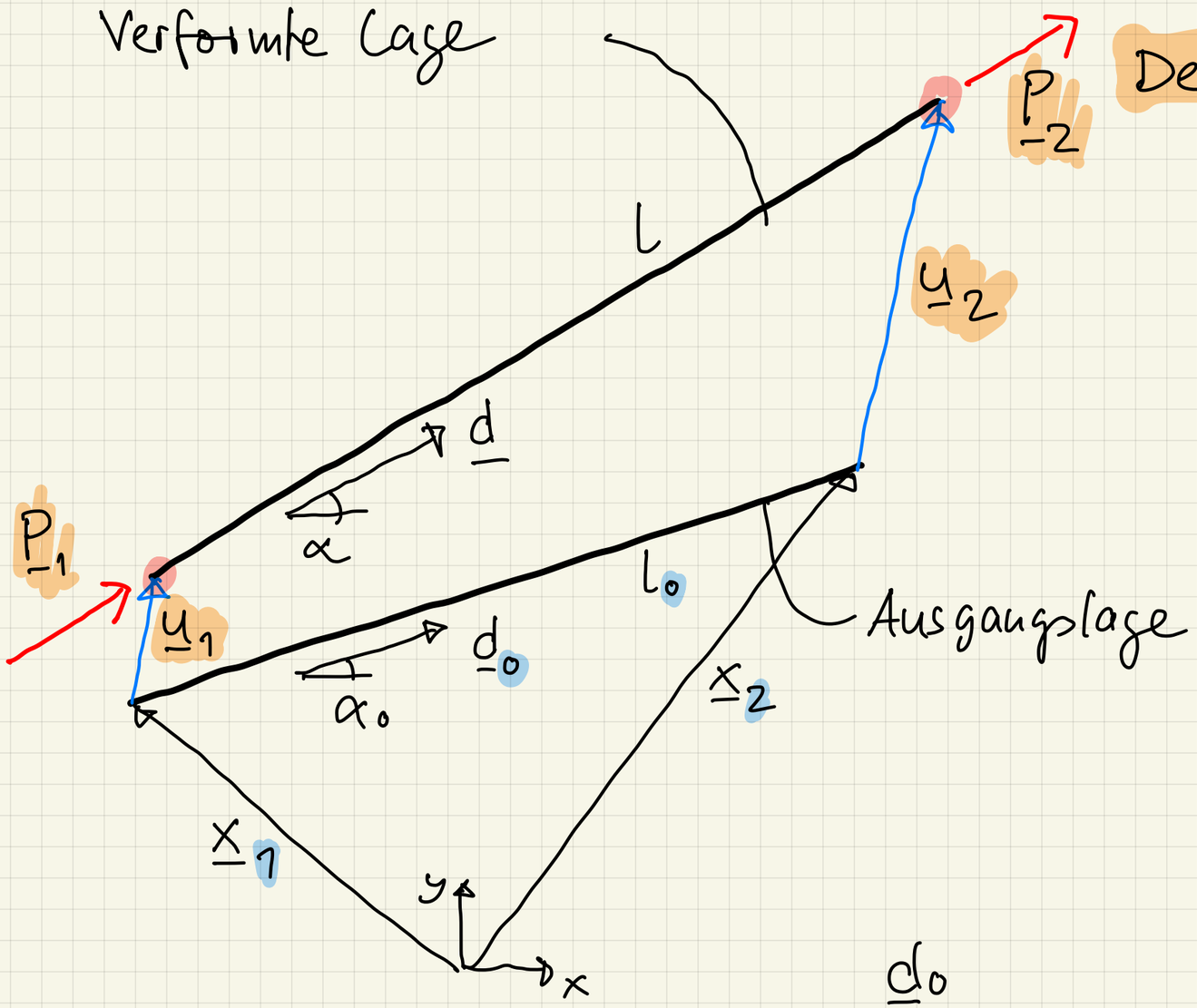
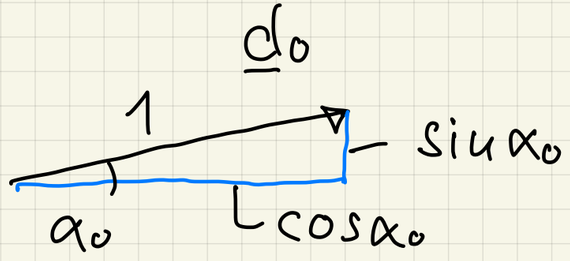
Finite Elemente für Stabtragwerke

Fachwerke : Stabendkräfte

Stabendkräfte Fachwerkstab 1/2





Definitionen

$$\underline{x}_1 = \begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix}, \quad \underline{x}_2 = \begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix}$$

$$\underline{d}_0 = \begin{pmatrix} d_{01} \\ d_{02} \end{pmatrix} = \begin{pmatrix} \cos\alpha_0 \\ \sin\alpha_0 \end{pmatrix} = \frac{1}{l_0}(\underline{x}_2 - \underline{x}_1), \quad |\underline{d}_0| = 1$$

$$l_0 = |\underline{x}_2 - \underline{x}_1| = \sqrt{(x_{21} - x_{11})^2 + (x_{22} - x_{12})^2}$$

$$\underline{u}_1 = \begin{pmatrix} u_{11} \\ u_{12} \end{pmatrix}, \quad \underline{u}_2 = \begin{pmatrix} u_{21} \\ u_{22} \end{pmatrix}$$

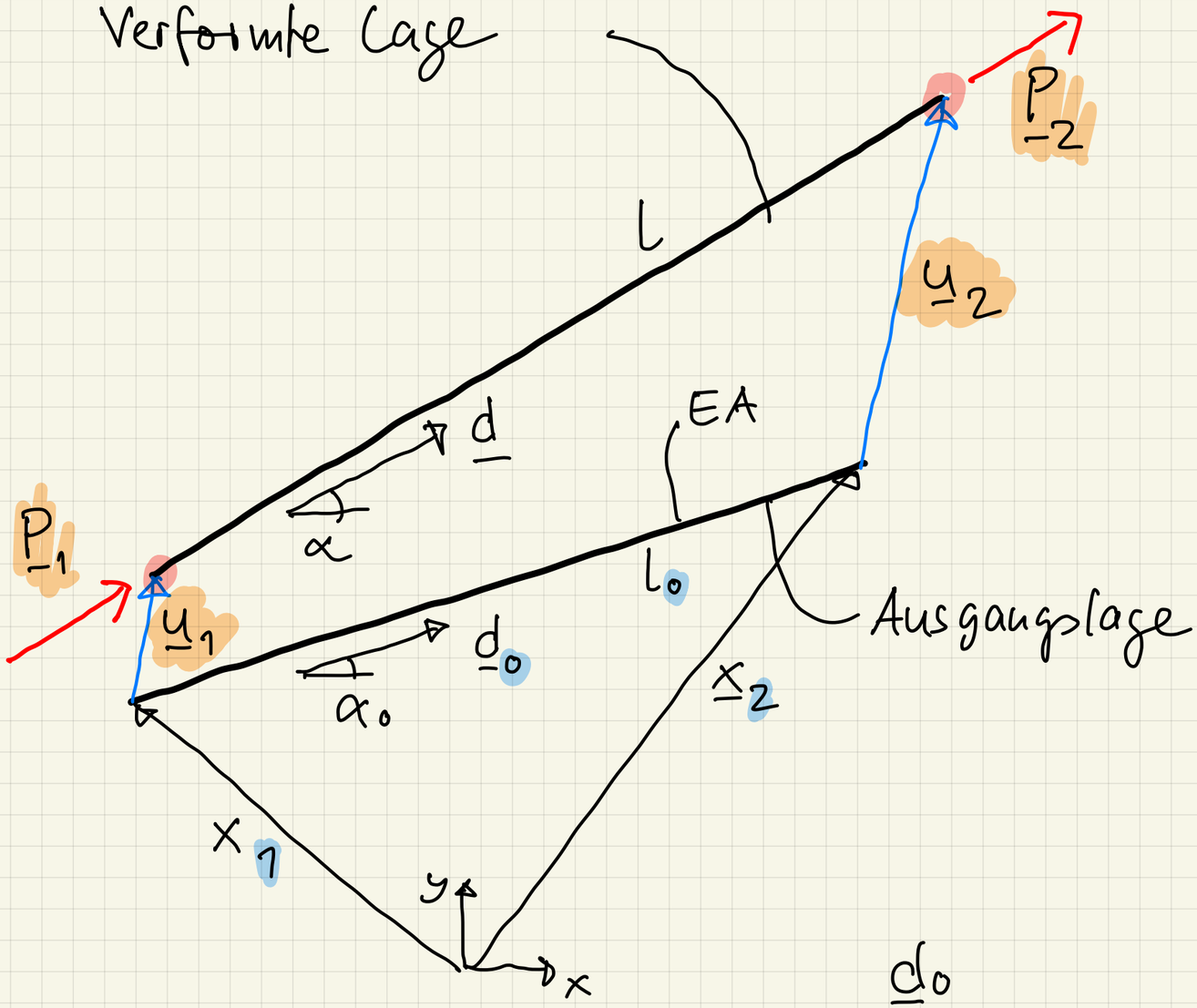
$$l = |\underline{x}_2 + \underline{u}_2 - (\underline{x}_1 + \underline{u}_1)| = \sqrt{\dots}$$

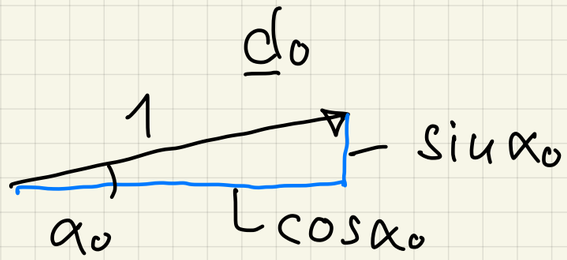
$$\underline{d} = \frac{1}{l}(\underline{x}_2 + \underline{u}_2 - (\underline{x}_1 + \underline{u}_1))$$

$$\underline{P}_1 = \begin{pmatrix} P_{11} \\ P_{12} \end{pmatrix}, \quad \underline{P}_2 = \begin{pmatrix} P_{21} \\ P_{22} \end{pmatrix}$$

$$\underline{P}_2 = -\underline{P}_1$$

Stabendkräfte Fachwerkstab 2/2





Gesucht

$$\underline{u} = \begin{pmatrix} \underline{u}_1 \\ \underline{u}_2 \end{pmatrix} = \begin{pmatrix} u_{11} \\ u_{12} \\ u_{21} \\ u_{22} \end{pmatrix}$$

$$\underline{P} = \begin{pmatrix} \underline{P}_1 \\ \underline{P}_2 \end{pmatrix} = \begin{pmatrix} P_{11} \\ P_{12} \\ P_{21} \\ P_{22} \end{pmatrix}$$

$$\underline{u} \in \mathbb{R}^4 \longrightarrow \boxed{\underline{P}} \longrightarrow \underline{P} \in \mathbb{R}^4$$

$$\underline{P} = \underline{P}(\underline{u})$$